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(54) SEAL TO BE DISPOSED BETWEEN A STRUCTURAL COMPONENT AND AN ENVIRONMENT MODULE

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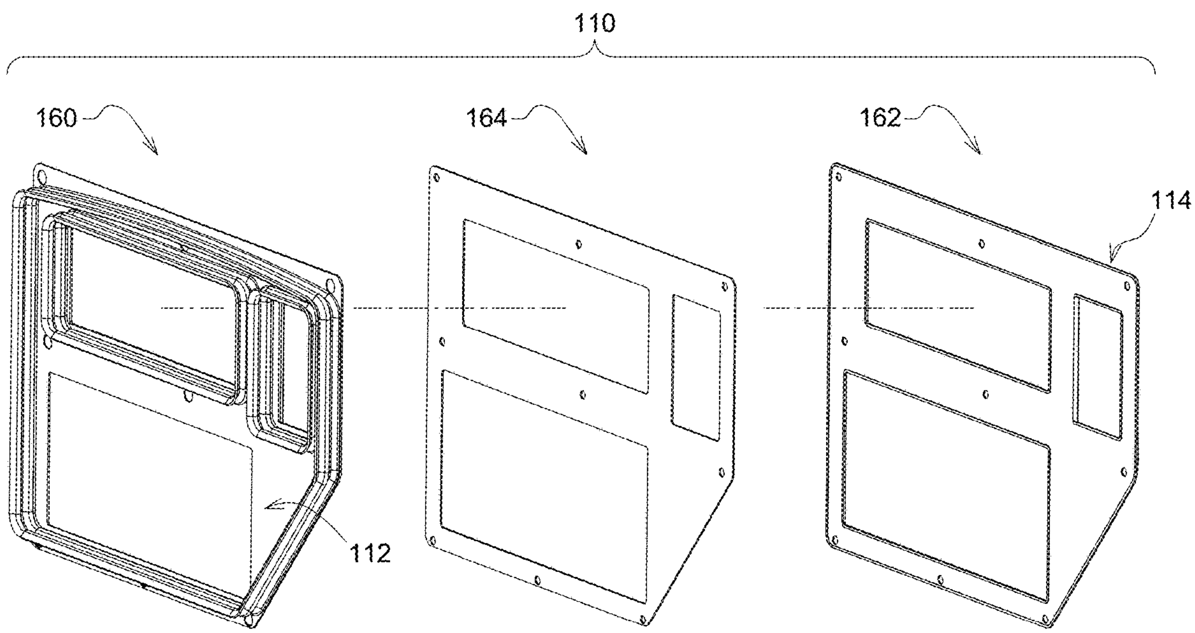
(57) **ABSTRACT**

A system includes a seal including a first opening disposed between a first portion of the structural component and a first portion of the environment module to create a negative pressure zone of the system and a second opening disposed between a second portion of the structural component and a second portion of the environment module to create a positive pressure zone of the system, the positive pressure zone surrounding the negative pressure zone.



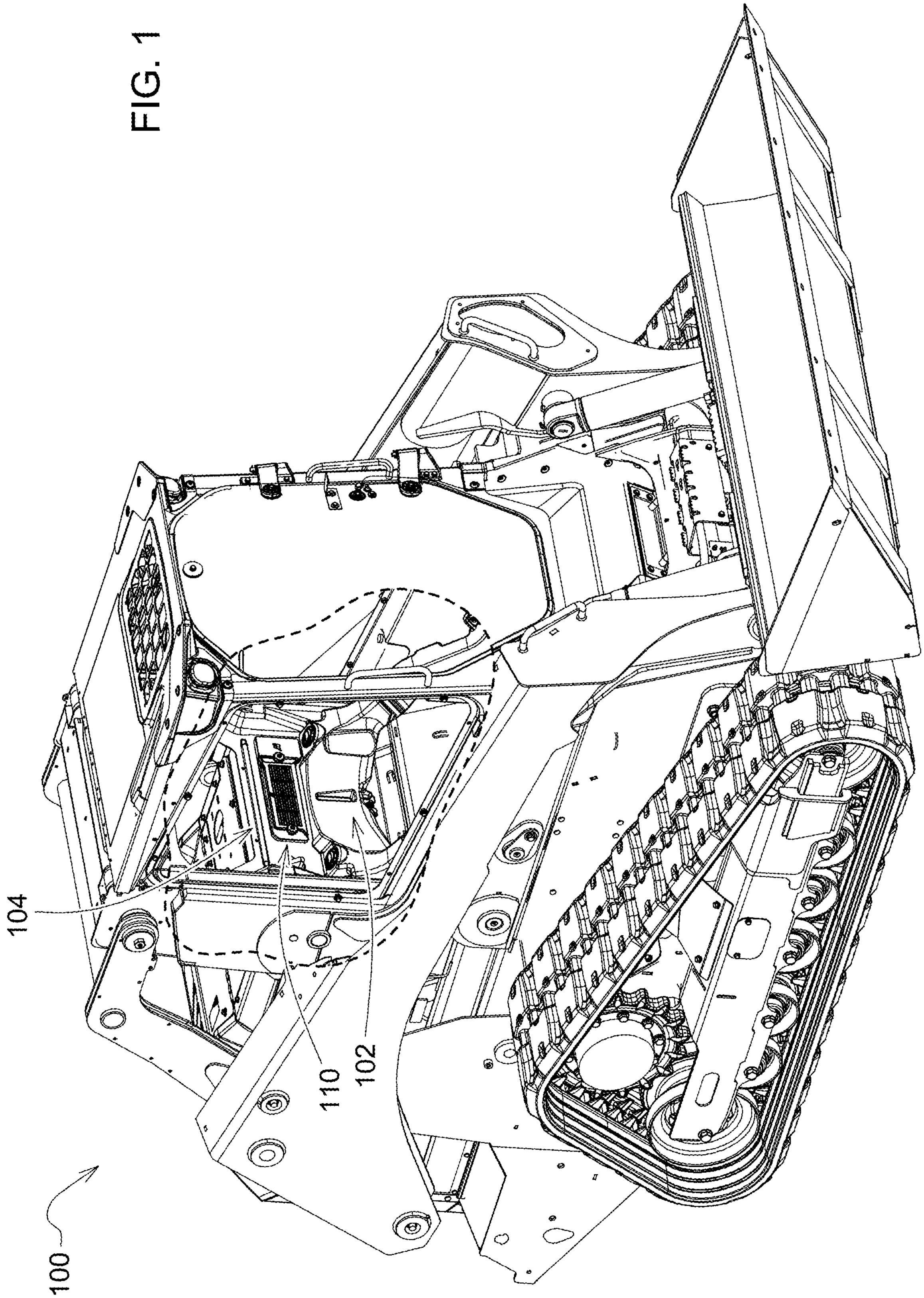


FIG. 1

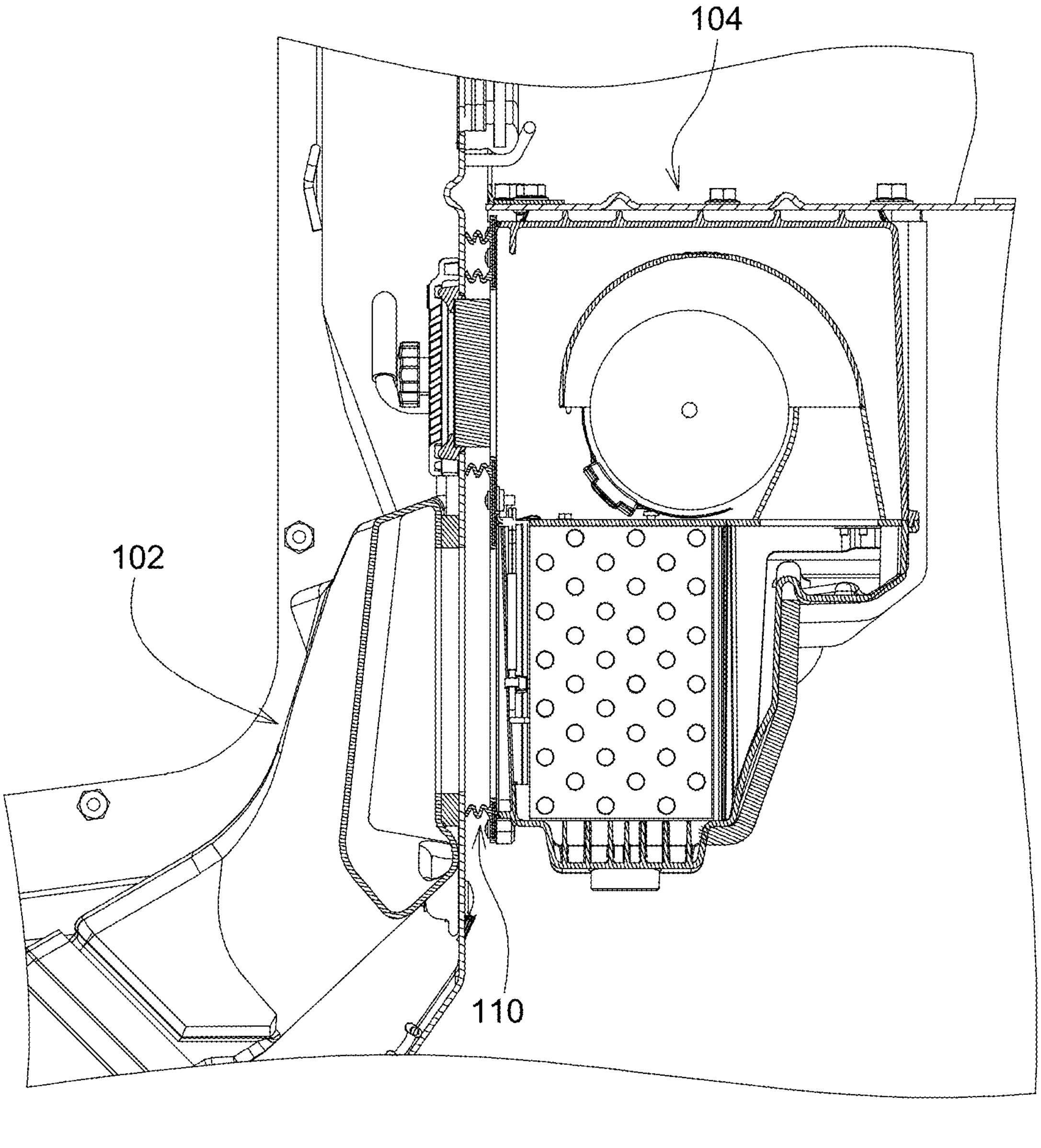


FIG. 2A

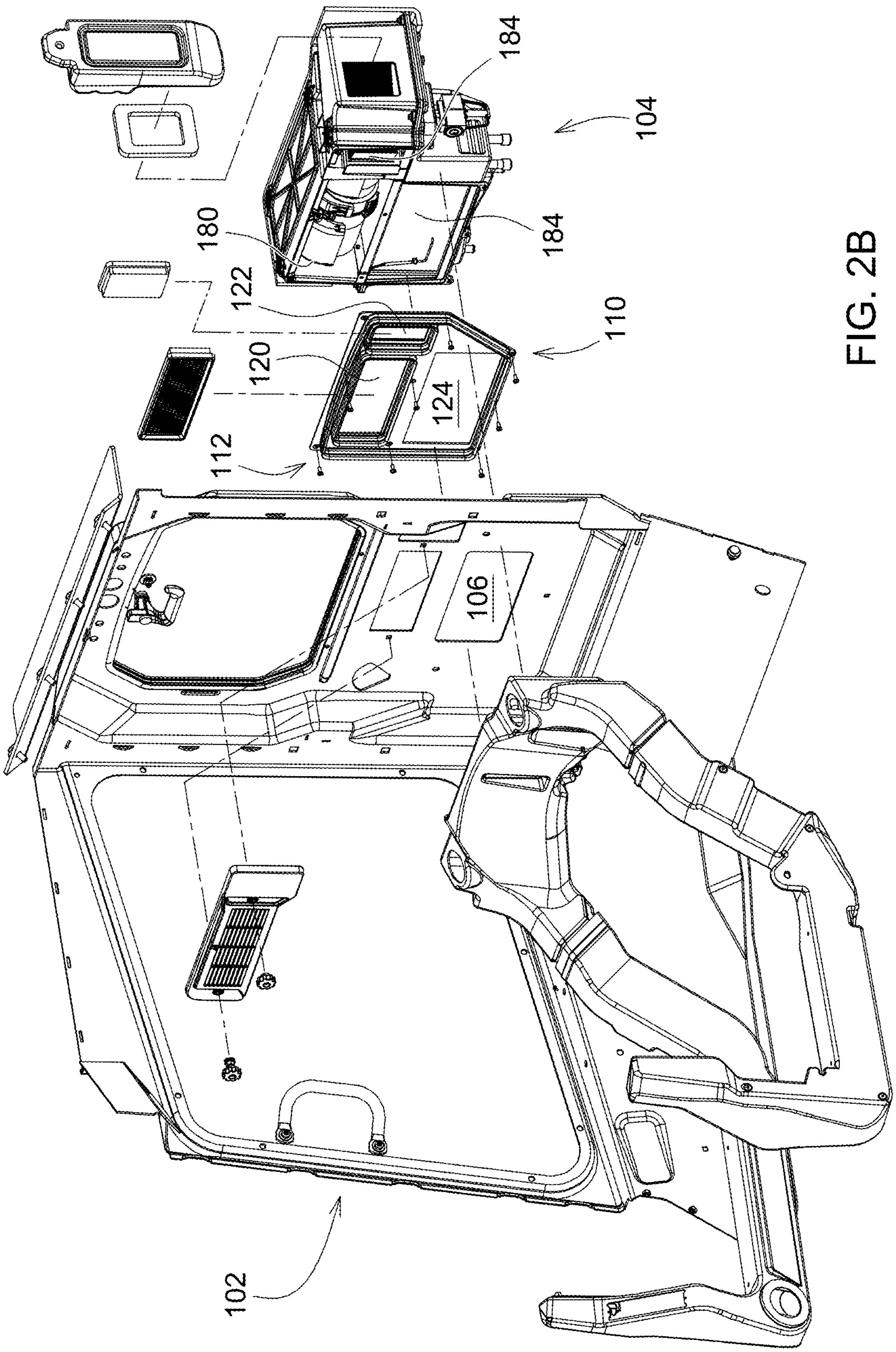


FIG. 2B

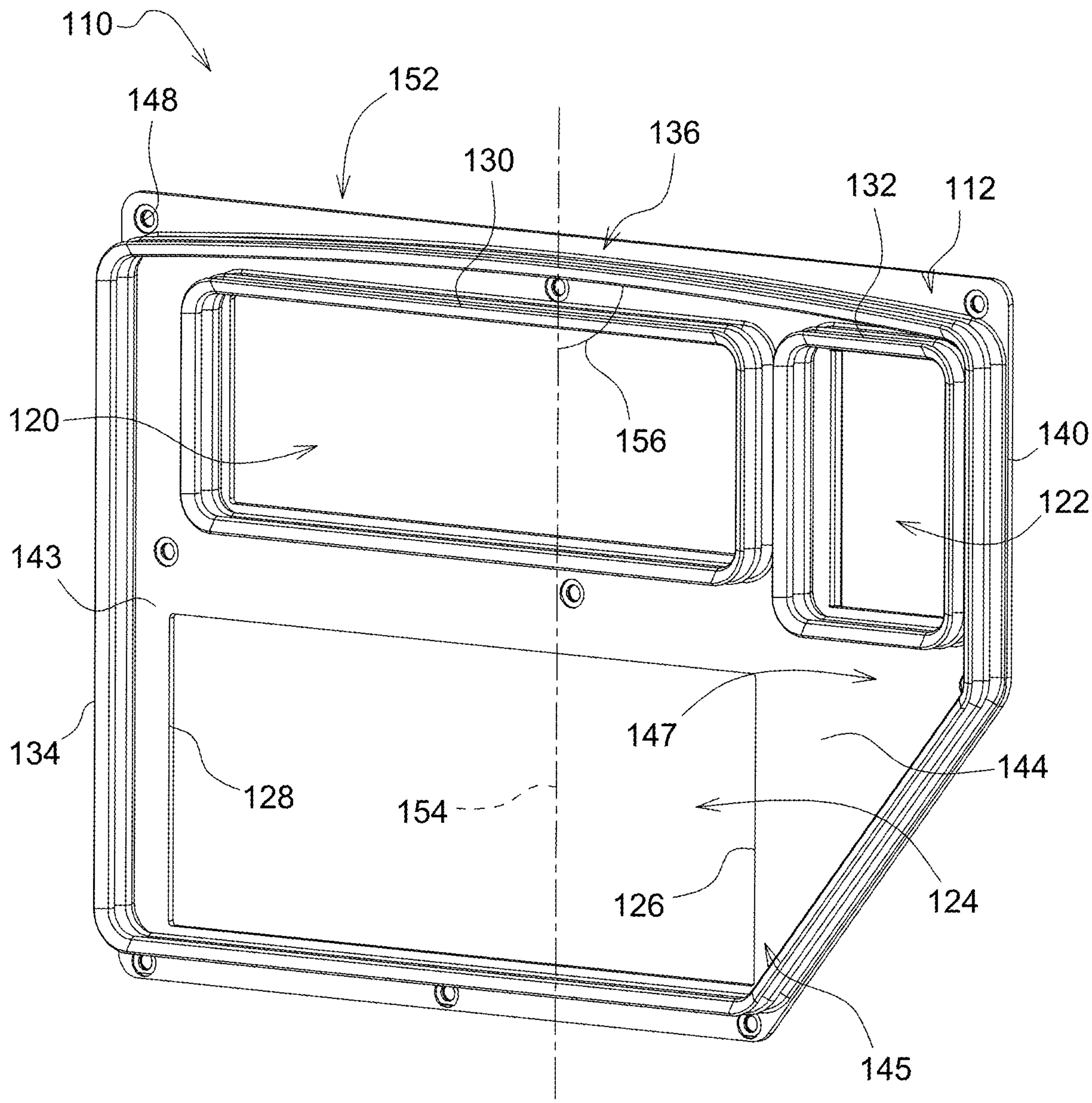

110
152
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FIG. 3A

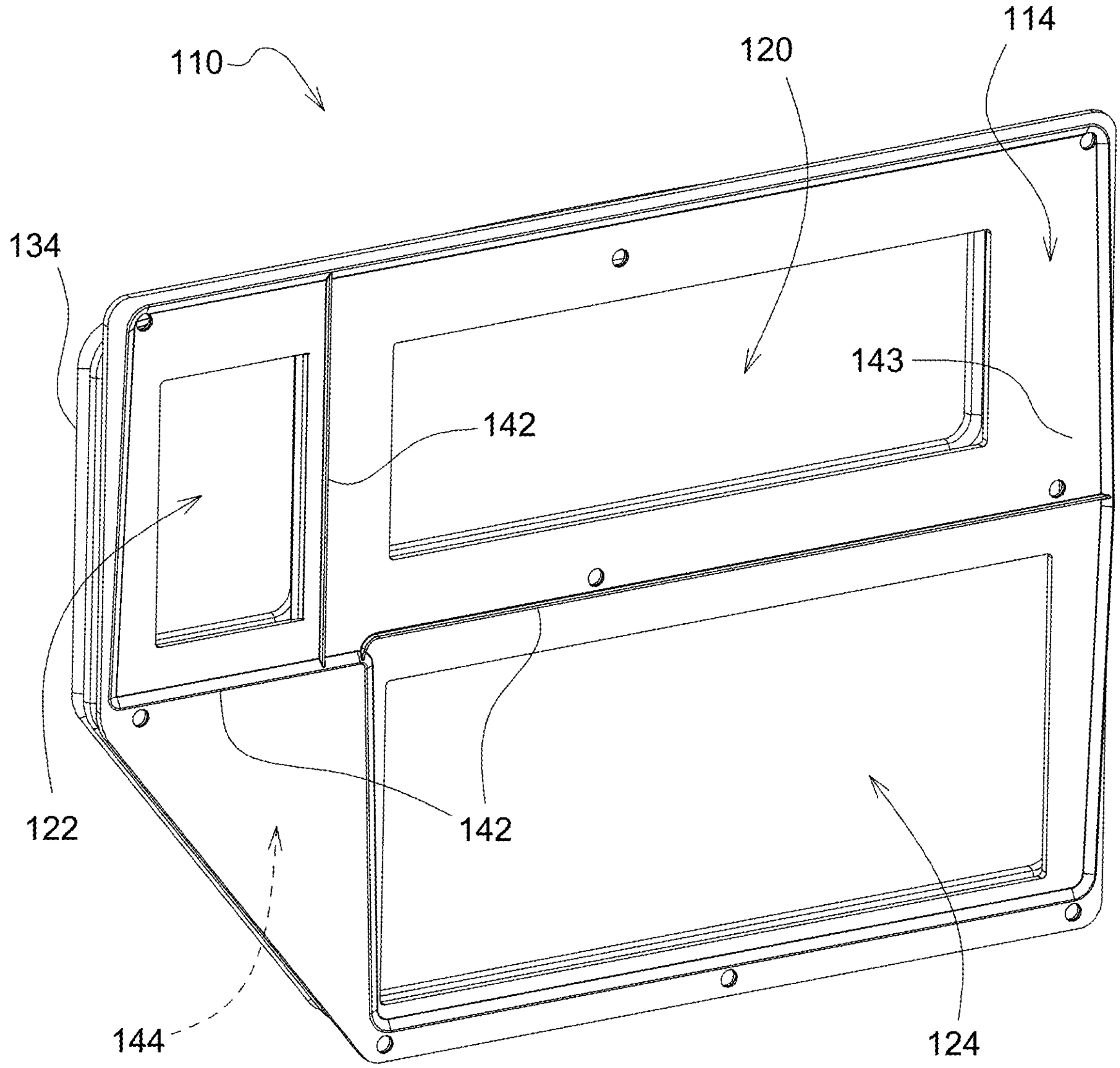

FIG. 3B

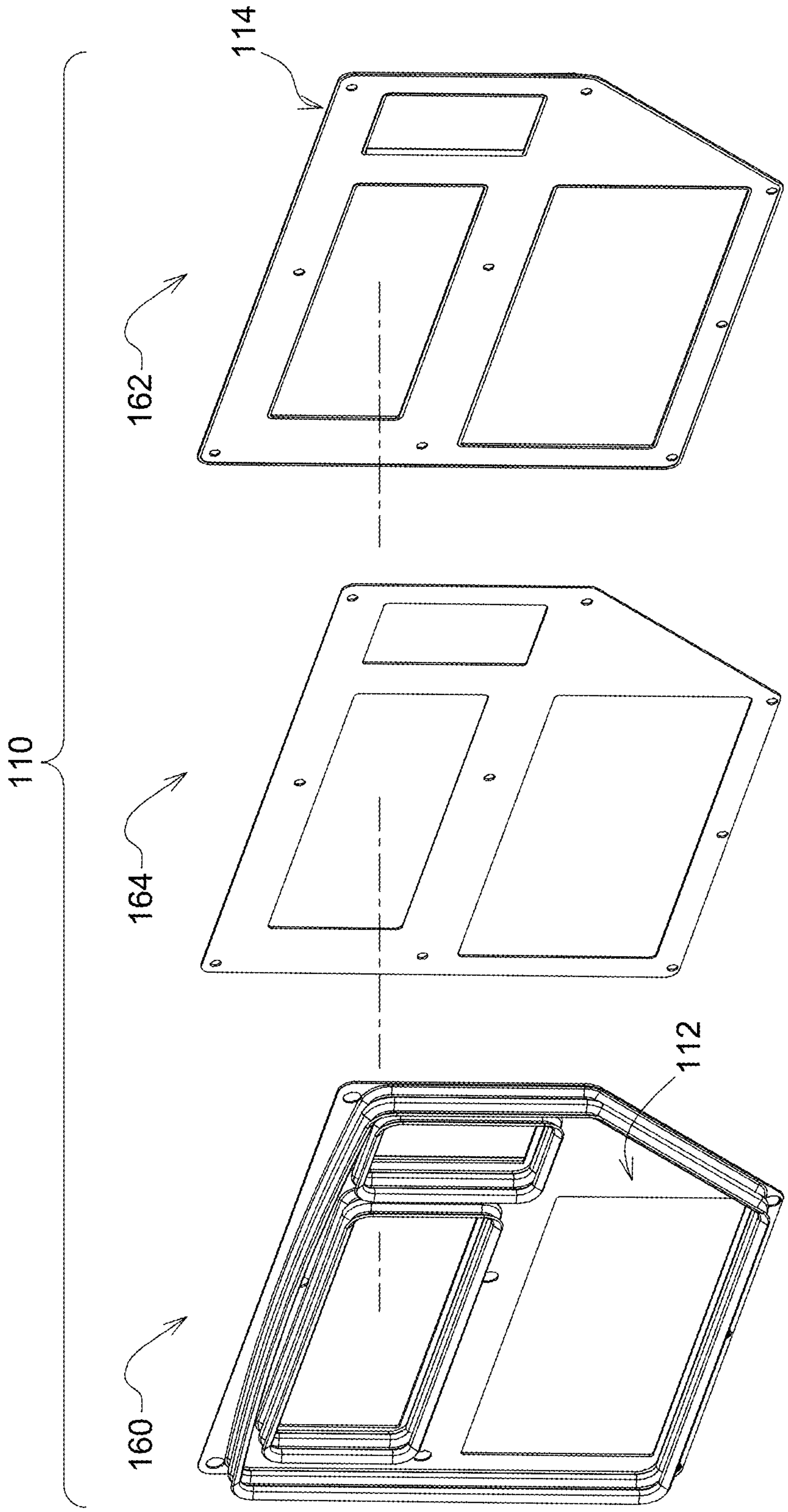

FIG. 3C

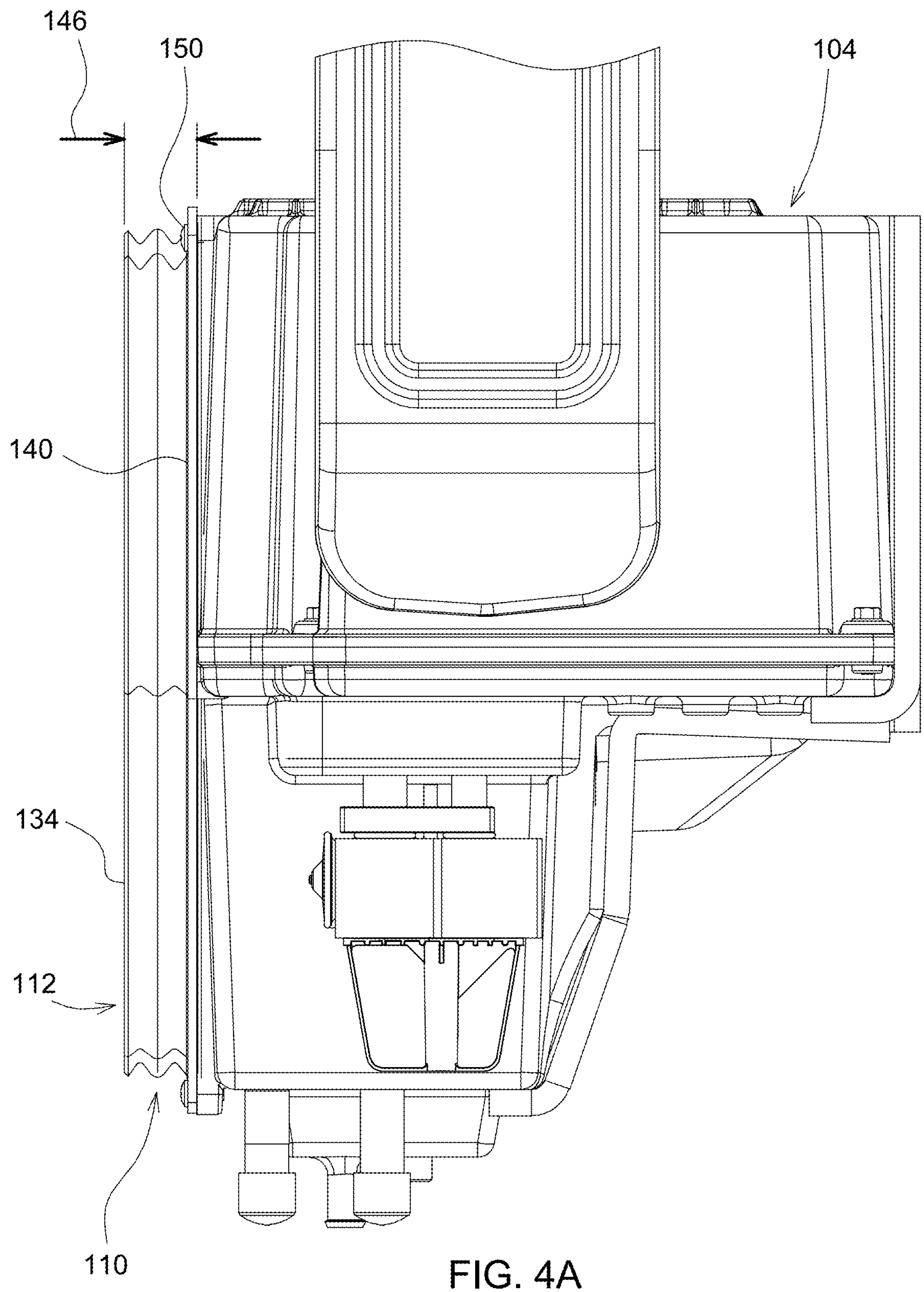

FIG. 4A

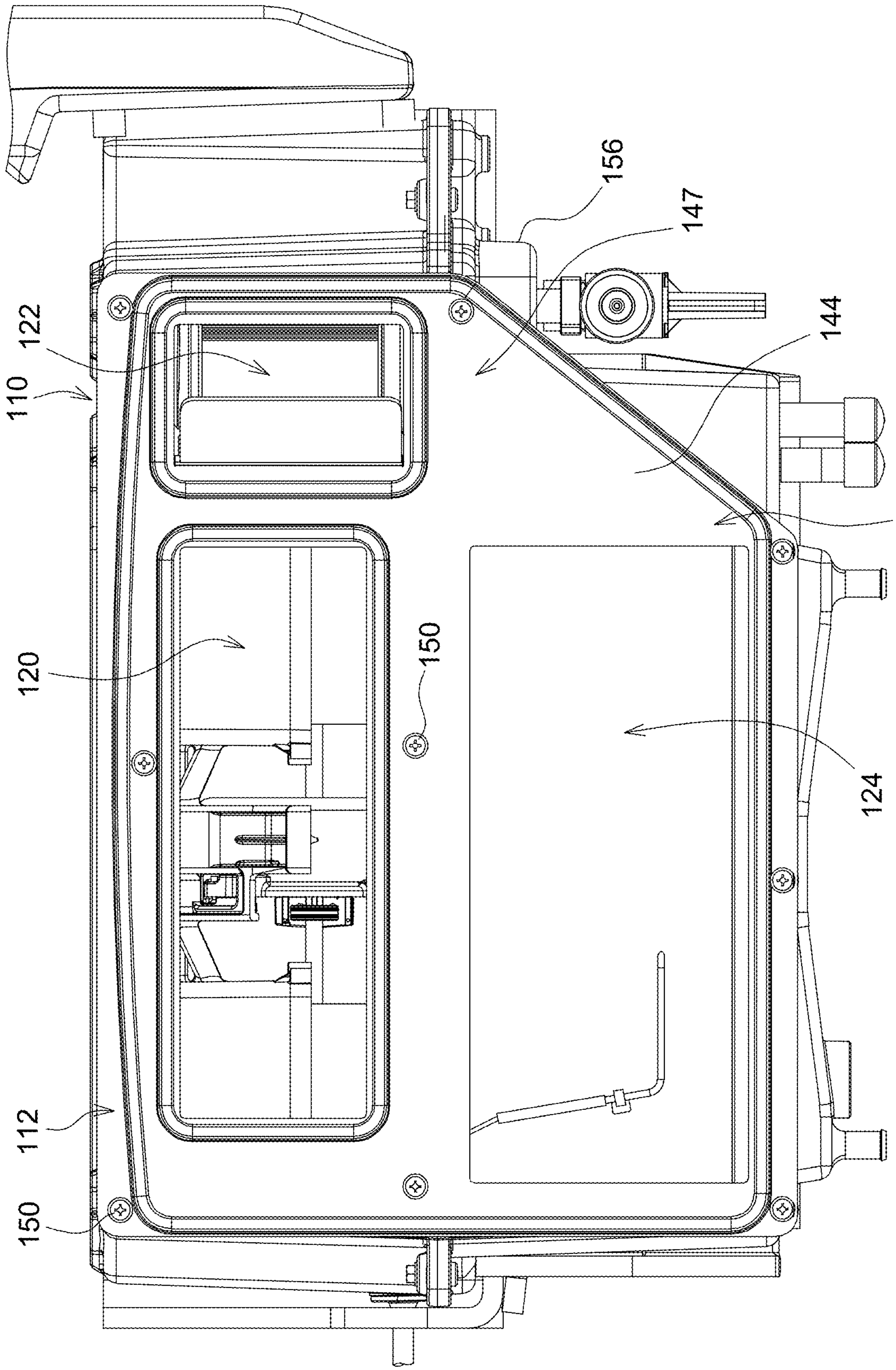

FIG. 4B

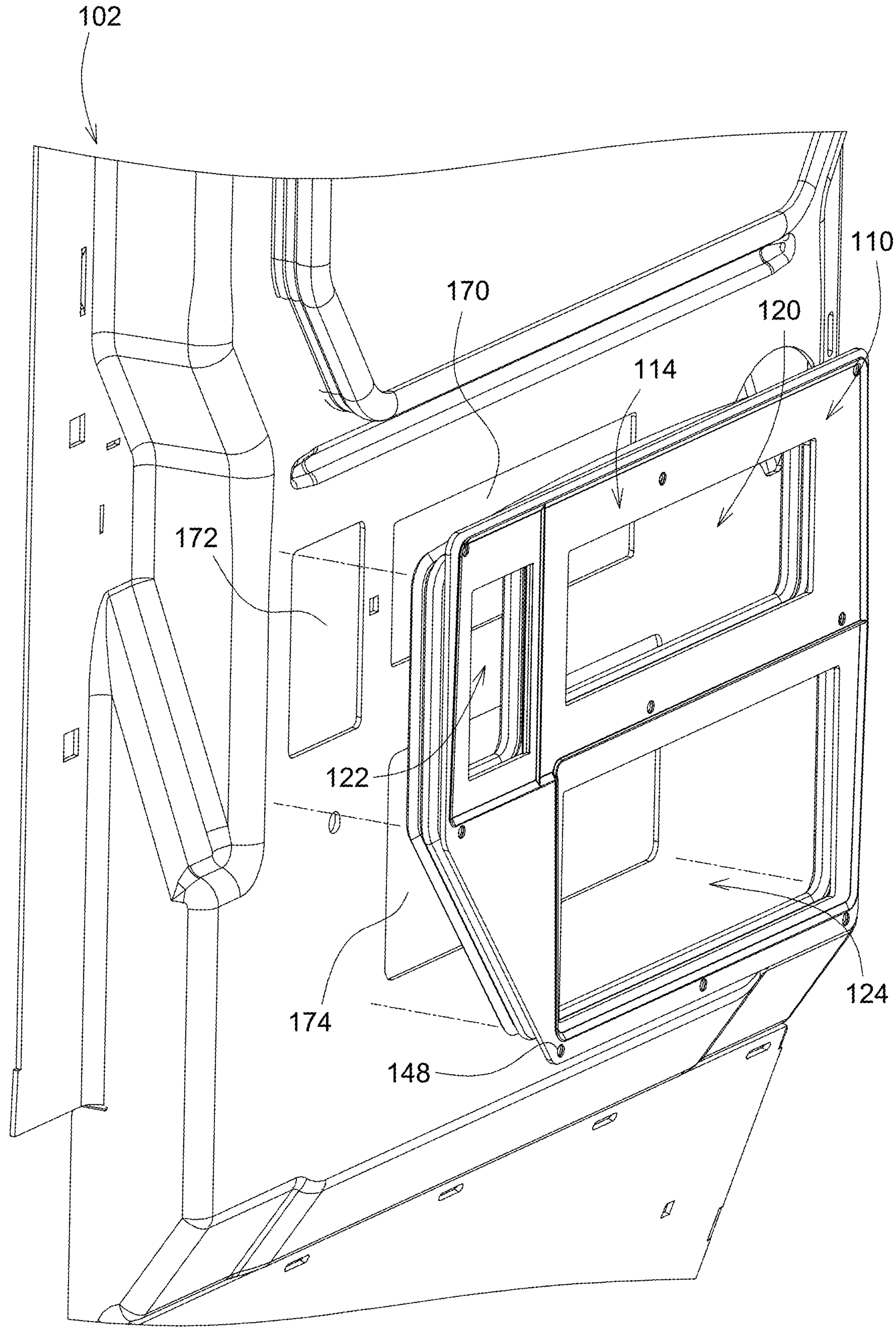

FIG. 5

SEAL TO BE DISPOSED BETWEEN A STRUCTURAL COMPONENT AND AN ENVIRONMENT MODULE

FIELD OF THE DISCLOSURE

[0001] Example embodiments relate to seals for placement between structural components (including, for example, vehicle cabs) and environment modules to be attached thereto.

BACKGROUND

[0002] Construction or utility vehicles (such as tractors) often have cabs that include environment modules (e.g., heating-ventilation-air conditioning units). Seals are often placed between the body of the cab and the environment module, for example, to limit water and/or dust ingress and to generally improve the performance of the environment module.

SUMMARY

[0003] In various aspects, the present disclosure provides an example seal for use between a structural component and an environment module attached thereto.

[0004] In at least one example embodiment, the seal may include a main body having opposing first and second sides and including one or more openings and a bellow structure extending from the first side of the main body, the bellow structure including a curvature relative to a major axis of the seal.

[0005] In at least one example embodiment, the bellow structure may be a continuous bellow structure extending along a perimeter of the first side.

[0006] In at least one example embodiment, the bellow structure may be a first bellow structure and the seal may further include a second bellow structure extending around a first opening of the one or more openings.

[0007] In at least one example embodiment, the seal may further include a third bellow structure extending around a second opening of the one or more openings.

[0008] In at least one example embodiment, the one or more openings may further include a third opening and a substantially planar portion of the main body may extend between a side of the third opening and the first bellow structure, a first end of the substantially planar portion having a first distance from the first bellow structure, and a second end of the substantially planar portion having a second distance from the first bellow structure, where the second distance is greater than the first distance.

[0009] In at least one example embodiment, the seal may include a metal plate defining the one or more openings, the metal plate being overmolded with a sealing material that defines the bellow structure.

[0010] In at least one example embodiment, the sealing material may have a durometer range greater than or equal to about 40 to less than or equal to about 70.

[0011] In at least one example embodiment, the seal may further include a plurality of coupling apertures extending through the main body.

[0012] In at least one example embodiment, the seal may have an average resting thickness greater than or equal to about 25 millimeters to less than or equal to about 40 millimeters.

[0013] In various aspects, the present disclosure provides another example seal for use between a structural component and an environment module attached thereto.

[0014] In at least one example embodiment, the seal may include a first opening surrounded or defined by a first bellow structure, a second opening surrounded or defined by a second bellow structure, and a third bellow structure surrounding the first and second bellow structures, the third bellow structure including a curvature relative to a major axis of the seal.

[0015] In at least one example embodiment, the seal may further include a third opening and a substantially planar portion of the seal extends between a side of the third opening and the first bellow structure, a first end of the substantially planar portion having a first distance from the first bellow structure, and a second end of the substantially planar portion having a second distance from the first bellow structure, where the second distance is greater than the first distance.

[0016] In at least one example embodiment, the seal may further include a fourth bellow structure surrounding or defining the third opening.

[0017] In at least one example embodiment, the seal may further include a fourth opening defining a positive pressure zone.

[0018] In at least one example embodiment, the seal may include a metal plate defining the first opening and the second opening, the metal plate being overmolded with a sealing material that defines the first, second, and third bellow structures.

[0019] In at least one example embodiment, the seal may have an average resting thickness greater than or equal to about 25 millimeters to less than or equal to about 40 millimeters.

[0020] In various aspects, the present disclosure provides another example seal for use between a structural component and an environment module attached thereto.

[0021] In at least one example embodiment, the seal may include one or more metal plates defining a main body having one or more openings and a sealing material overmolding the main body and extending from one side thereof to form a bellow structure, the bellow structure including a curvature relative to a major axis of the seal.

[0022] In at least one example embodiment, the bellow structure may extend along a perimeter of the one side of the main body and the seal may further include a second bellow structure extending around a first opening of the one or more openings.

[0023] In at least one example embodiment, the seal may further include a third bellow structure extending around a second opening of the one or more openings.

[0024] In at least one example embodiment, the seal may have an average resting thickness greater than or equal to about 25 millimeters to less than or equal to about 40 millimeters.

[0025] In various aspects, the present disclosure provides a system having a structural component and an environment module.

[0026] In at least one example embodiment, the system includes a seal including a first opening disposed between a first portion of the structural component and a first portion of the environment module to create a negative pressure zone of the system and a second opening disposed between a second portion of the structural component and a second

portion of the environment module to create a positive pressure zone of the system, the positive pressure zone surrounding the negative pressure zone.

BRIEF DESCRIPTION OF THE DRAWINGS

[0027] The various features and advantages of the non-limiting embodiments herein may become more apparent upon review of the detailed description in conjunction with the accompanying drawings. The accompanying drawings are merely provided for illustrative purposes and should not be interpreted to limit the scope of the claims. The accompanying drawings are not to be considered as drawn to scale unless explicitly noted. For the purposes of clarity, various dimensions of the drawings may have been exaggerated.

[0028] FIG. **1** is a perspective view of an example vehicle including a seal disposed between an environment module and a cab of a vehicle in accordance with at least one example embodiment of the present disclosure.

[0029] FIG. **2**A is a partial perspective side view of the vehicle of FIG. **1** showing the seal as disposed between the environment module and the cab of the vehicle in accordance with at least one example embodiment of the present disclosure.

[0030] FIG. **2**B is an exploded view of FIG. **2**A in accordance with at least one example embodiment of the present disclosure.

[0031] FIG. **3**A is a front perspective view of the seal illustrated in FIGS. **1**-**2**B in accordance with at least one example embodiment of the present disclosure.

[0032] FIG. **3**B is a rear perspective view of the seal illustrated in FIGS. **1**-**3**A in accordance with at least one example embodiment of the present disclosure.

[0033] FIG. **3**C is an exploded view of the seal illustrated in FIGS. **1**-**3**B in accordance with at least one example embodiment of the present disclosure.

[0034] FIG. **4**A is a side view of the seal illustrated in FIGS. **1**-**3**C as coupled to the environment module in accordance with at least one example embodiment of the present disclosure.

[0035] FIG. **4**B is a front view of the seal illustrated in FIGS. **1**-**4**A as coupled to the environment module in accordance with at least one example embodiment of the present disclosure.

[0036] FIG. **5** is a rear view of the seal illustrated in FIGS. **1**-**4**B as coupled to the cap in accordance with at least one example embodiment of the present disclosure.

DETAILED DESCRIPTION

[0037] Some detailed example embodiments are disclosed herein. However, specific structural and functional details disclosed herein are merely representative for the purposes of describing example embodiments. Example embodiments may, however, be embodied in many alternate forms and should not be construed as limited to only the example embodiments set forth herein.

[0038] Accordingly, while example embodiments are capable of various modifications and alternative forms, example embodiments thereof are shown by way of example in the drawings and will herein be described in detail. It should be understood, however, that there is no intent to limit example embodiments to the particular forms disclosed, but to the contrary, example embodiments are to cover all modifications, equivalents, and alternatives falling within the scope of example embodiments. Like numbers refer to like elements throughout the description of the figures.

[0039] It should be understood that when an element or layer is referred to as being "on," "connected to," "coupled to," or "covering" another element or layer, it may be directly on, connected to, coupled to, or covering the other element or layer or intervening elements or layers may be present. In contrast, when an element is referred to as being "directly on," "directly connected to," or "directly coupled to" another element or layer, there are no intervening elements or layers present. Like numbers refer to like elements throughout the specification. As used herein, the term "and/or" includes any and all combinations of one or more of the associated listed items.

[0040] It should be understood that, although the terms first, second, third, etc. may be used herein to describe various elements, regions, layers and/or sections, these elements, regions, layers, and/or sections should not be limited by these terms. These terms are only used to distinguish one element, region, layer, or section from another region, layer, or section. Thus, a first element, component, region, layer, or section discussed below could be termed a second element, region, layer, or section without departing from the teachings of example embodiments.

[0041] Spatially relative terms (e.g., "beneath," "below," "lower," "above," "upper," and the like) may be used herein for ease of description to describe one element or feature's relationship to another element(s) or feature(s) as illustrated in the figures. It should be understood that the spatially relative terms are intended to encompass different orientations of the device in use or operation in addition to the orientation depicted in the figures. For example, if the device in the figures is turned over, elements described as "below" or "beneath" other elements or features would then be oriented "above" the other elements or features. Thus, the term "below" may encompass both an orientation of above and below. The device may be otherwise oriented (rotated 90 degrees or at other orientations) and the spatially relative descriptors used herein interpreted accordingly.

[0042] The terminology used herein is for the purpose of describing various example embodiments only and is not intended to be limiting of example embodiments. As used herein, the singular forms "a," "an," and "the" are intended to include the plural forms as well, unless the context clearly indicates otherwise. It will be further understood that the terms "includes," "including," "comprises," and/or "comprising," specify the presence of stated features, integers, steps, operations, and/or elements, but do not preclude the presence or addition of one or more other features, integers, steps, operations, elements, and/or groups thereof.

[0043] When the terms "about" or "substantially" are used in this specification in connection with a numerical value, it is intended that the associated numerical value includes a manufacturing or operational tolerance (e.g., ±10%) around the stated numerical value. Moreover, when the terms "generally" or "substantially" are used in connection with geometric shapes, it is intended that precision of the geometric shape is not required but that latitude for the shape is within the scope of the disclosure. Furthermore, regardless of whether numerical values or shapes are modified as "about," "generally," or "substantially," it will be understood that these values and shapes should be construed as including a manufacturing or operational tolerance (e.g., ±10%) around the stated numerical values or shapes.

[0044] Unless otherwise defined, all terms (including technical and scientific terms) used herein have the same meaning as commonly understood by one of ordinary skill in the art to which example embodiments belong. It will be further understood that terms, including those defined in commonly used dictionaries, should be interpreted as having a meaning that is consistent with their meaning in the context of the relevant art and will not be interpreted in an idealized or overly formal sense unless expressly so defined herein.

[0045] At least some example embodiments of the present disclosure provide seals for use between structural components (including, for example, vehicle cabs) and environment modules (e.g., heating-ventilation-air condition unit) attached thereto.

[0046] As used herein, vehicle may include, for example, an automotive vehicle and/or an agricultural vehicle and/or a mining vehicle and/or a construction vehicle and/or a utility vehicle. The vehicle may include, for example, backhoe loaders, tractors, planters, sprayers, bulldozers, forklifts, steam rollers, cranes, haul truck, underground graders, rock breakers, and the like.

[0047] FIG. **1** is a perspective view of an example vehicle **100** including a seal **110** in accordance with at least one example embodiment of the present disclosure. The seal **110** is disposed between the cab **102** of the vehicle **100** and an environment module **104** attached thereto. The seal **110** is a dynamic seal configured to maintain an appropriate seal between the cab **102** and the environment module **104** during relative movement of the cab **102** and/or environment module **104**. Although the vehicle **100** is illustrated as including only a single environment module **104** and a single seal **110** associated therewith, it should be appreciated that, in other example embodiments, vehicles may include two or more environment modules and related seals.

[0048] FIG. **2**A is a partial perspective side view of the vehicle of FIG. **1**, and FIG. **2**B is an exploded view of the partial perspective side view of FIG. **2**A. As illustrated, a first (or front) side (or surface) **112** of the seal **110** is aligned with and interfaces with the cab **102** (and more particularly, with one or more apertures **106** in the cab **102** configured to join with the environment module **104**), while a second (or rear) side (or surface) **114** of the seal **110** is aligned with and interfaces with (e.g., couples to) the environment module **104**. As illustrated, the seal **110** includes a plurality of openings **120**, **122**, **124** configured to align with different components of the cab **102** and/or environment module **104**. For example, in at least one example embodiment, the seal **110** may include three openings: a first opening **120**, a second opening **122**, and a third opening **124**. The first opening **120** of the seal **110** may correspond with respective first sections (or portions) of the cab **102** and environment module **104**, the second opening **122** of the seal **110** may correspond with respective second sections (or portions) of the cab **102** and environment module **104**, and the third opening **124** of the seal **110** may correspond with respective third sections (or portions) of the cab **102** and environment module **104**.

[0049] In at least one example embodiment, the first opening **120** of the seal **110** may be configured (e.g., shaped) to align and interface with (i.e., seal against) a cabin recirculation air portion (or section) **170** of the cab **102** and/or a cabin recirculation air portion (or section) **180** of the environment module **104**. In at least one example embodiment, the second opening **122** of the seal **110** may be configured (e.g., shaped) to align and interface with (i.e., seal against) a cabin fresh air portion (or section) **172** of the cab **102** and/or a cabin fresh air portion (or section) **182** of the environment module **104**. In at least one example embodiment, the third opening **124** of the seal **110** may be configured e.g., shaped) to align and interface with (i.e., seal against) a HVAC module outlet or supply air portion (or section) **174** of the cab **102** and/or a HVAC module outlet or supply air portion (or section) **184** of the environment module **104**.

[0050] In at least one example embodiment, the first opening **120** of the seal **110** may define a negative pressure region (or portion or zone) relation relative to the outside environment. The negative pressure may be caused by the cabin recirculation air portion (or section) **170** of the cab **102** and/or the cabin recirculation air portion (or section) **180** of the environment module **104**. In at least one example embodiment, the second opening **122** of the seal **110** may define another negative pressure region (or portion or zone) relative to the outside environment. The negative pressure may be caused by the cabin fresh air portion (or section) **172** of the cab **102** and/or the cabin fresh air portion (or section) **182** of the environment module **104**. In at least one example embodiment, the third opening **124** of the seal may define a positive pressure region (or portion or zone) relative to the outside environment. The positive pressure may be caused by the HVAC module outlet or supply air portion (or section) **174** of the cab **102** and/or the HVAC module outlet or supply air portion (or section) **184** of the environment module **104**. In at least one example embodiment, the positive pressure at third opening **123** provides a positive pressure zone that helps to prevent outside environmental air and/or contaminants (e.g., dust, debris) from reaching the first and second openings **120**, **122**, for example, in the instance of a momentary, or small, lose of the sealing between the cab **102** and the environment module **104** such as in the instances of rapid movement of the cab **102** and/or environment module **104**.

[0051] Although three openings are illustrated, it should be appreciated that, in various other example embodiment the seal **110** may include one or more openings. For example, in at least one example embodiment, the seal may include four openings, one of the four openings being a communication port having (or defining), for example, a positive pressure, while the three other openings have a negative or neutral pressure.

[0052] FIG. **3**A is a perspective view of the front surface **112** of the seal **110**. One or more bellow structures may extend from the front surface **112** of the seal. The one or more bellow structures may be continuous or discontinuous. For example, in at least one example embodiment, as illustrated, a first bellow structure **130** may be a continuous structure that extends from (or defines) portion of the front surface **112** surrounding (or defining) the first opening **120** of the seal **110**. In at least one example embodiment, the first bellow structure **130** may be integral to the first opening **120** of the seal **110**. In each instance, the first bellow structure **130** provides for flexible sealing around and between a first section of the cab **102** and a first section of the environment module **104** such that movement of the cab **102** and environment module **104** are readily absorbed and tolerated.

[0053] In at least one example embodiment, as illustrated, a second bellow structure **132** may be a continuous structure that extends from (or define) a portion of the front surface **112** surrounding (or defining) the second opening **122** of the seal **110**. In at least one example embodiment, the second bellow structure **132** may be integral to the second opening **122** of the seal **110**. In each instance, the second below structure **132** provides for flexible sealing around and between a second portion of the cab **102** and a second portion of the environment module **104** such that movement of the cab **102** and environment module **104** are readily absorbed and tolerated.

[0054] In at least one example embodiment, as illustrated, a third bellow structure **134** may be a continuous structure that extend from (or define) at least a portion of a perimeter **140** of the front surface **112** of the seal **110**. The third bellow structure **134** provides for flexible sealing around and between the cab **102** and the perimeter **140** of the environment module **104** such that movement of the cab **102** and environment module **104** are readily absorbed and tolerated.

[0055] In at least one example embodiment, as illustrated, the third bellow structure **134** is constructed (or configured) to direct (or shed) possible contaminants (e.g., water) around the same and away from the sealed area. For example, the third bellow structure **134** may include a crown portion **136** having a slight curvature **156** relative to a major axis **154** of the seal **110**. The crown portion **136** may be disposed near (or aligned with) a first (or top) portion **152** of the seal **110**. The curvature **156** is selected to limit or eliminate low spots where water (or other fluids) may pool regardless of the slope at which the vehicle **100** may be parked and accordingly the angle of the cab **102** and/or environment module **104**. The curvature **156** is selected to direct the water (or other fluid) in an assigned direction.

[0056] In at least one example embodiment, as illustrated, the third opening **124** may be free of a bellows. The third opening **124** may have a first side **126** and an opposing second side **128**. A first distance between the first side **126** of the third opening **124** and the third bellow structure **134** may be greater than a second distance between the second side **128** of the third opening **124** and the third bellow structure **134**. In at least one example embodiment, as illustrated in FIG. **3**A, a substantially planar portion **144** of a main body **143** of the seal **110** may extend between the first side **126** of the third opening **124** and the third bellow structure **134**. The perimeter **140** of the seal may be shaped such that a first end **145** of the substantially planar portion **144** has a first distance from the third bellow structure **134**, and a second end **147** of the substantially planar portion **144** has a second distance from the third bellow structure **134**. The second distance may be greater than the first distance. In at least one example embodiment, the substantially planar portion may have a general triangular configuration.

[0057] Although not illustrated, it should be appreciated that, in at least one example embodiment, another bellow structure may surround or define the third opening **124** of the seal **110**. Similar to the first bellow structure **130** and/or the second bellow structure **132**, the another bellow structure may extend alone (or define) a wall portion of the seal **110** defining the third opening **124**. The another bellow structure may provide for flexible sealing around and between a third portion of the cab **102** and a third portion of the environment module **104** such that movement of the cab **102** and environment module **104** are readily absorbed and tolerated.

[0058] Although three bellow structure are illustrated, it should be appreciated that, in at least one example embodiment, the seal **110** may include only one of the first bellow structure **130**, the second bellow structure **132**, the third bellow structure **134**, and the another bellow structure. Likewise, it should be appreciated that, in at least one example embodiment, the seal **110** may include two of the first bellow structure **130**, the second bellow structure **132**, the third bellow structure **134**, and the another bellow structure (e.g., the first bellow structure **130** and the second bellow structure **132** or the first bellow structure **130** and the third bellow structure **134** or the first bellow structure **130** and the another bellow structure or the second bellow structure **132** and the third bellow structure **134** or the second bellow structure **132** and the another bellow structure or the third bellow structure **134** and the another bellow structure).

[0059] Although the bellow structures as illustrated are continuous structures, it should be appreciated that, in at least one example embodiment, one or more of the different bellow structures (e.g., the first bellow structure **130** and/or the second bellow structure **132** and/or the third bellow structures **134** and/or the another bellow structure) may be non-continuous structures.

[0060] FIG. **3**B is a perspective view of the rear surface **114** of the seal **110**. As illustrated, in at least one example embodiment, the rear surface **114** of the seal **110** may be generally planar that provides a generally smooth and consistent sealing surface (see, e.g., FIG. **4**A). The rear surface **114** may include one or more transition regions **142** that disposed between the different portions the seal **110** including the first, second, and third openings **120**, **122**, **124**, respectively. These transitions regions may provide texturization that helps to improve the sealing between the different portions of the cab **102** and the different portions of the environment module **104**.

[0061] FIG. **3**C is an exploded view of the seal **110**. In at least one example embodiment, the seal **110** may include one or more metal plates having or defining one or more openings or apertures (e.g., defining the first, second, and third openings **120**, **122**, **134**) that are overmolded with a sealing material. For example, as illustrated, the seal **110** may include a first half **160** that defines the first side **112** of the seal **110**, a second half **162** that defines the second side **114** of the seal **110**, and a metal plate **164** disposed therebetween. The first and second halves **160**, **162** may include one or more sealing materials and may join together to encase the metal plate **164**. In at least one example embodiment, sealing material may have a durometer range greater than or equal to about 40 to less than or equal to about 70. For example, the sealing material may include ethylene propylene diene monomer (EPDM) rubber.

[0062] Although only a single metal plate **164** is illustrated in FIG. **3**C, it should be appreciated that, in various other example embodiments, one or more metal plates may be encased by the first and second halves **160**, **162**. The use (or incorporation) of the one or more metal plates **164** helps to ensure a smooth and consistent sealing surface for engaging and interacting with the environment module **104** (see, e.g., FIG. **4**A).

[0063] FIG. **4**A is a side view of the rear surface **114** of the seal **110** as coupled to the environment module **104**. As illustrated, the rear surface **114** is substantially flushed with the corresponding surface of the environment module **104**. As illustrated, in at least one example embodiment, the seal **110** may have an average thickness **146** (before compression of the environment module **104** and the seal **110** against the cab **102**) of greater than or equal to about 25 millimeters to less than or equal to about 40 millimeters, and optionally, about 32 millimeters. In at least one example embodiment, each of the bellow structures **130**, **132**, **134** may have a height extending from the planar surface of the seal **110** of greater than or equal to about 15 millimeters to less than or equal to about 35 millimeters, and optionally, about 23 millimeters.

[0064] FIG. **4**B is a front view of the rear surface **114** of the seal **110** as coupled to the environment module **104**. As illustrated, the seal **110** has an overall size greater than the corresponding parts (to be sealed) of the cab **102** and/or environment module **104**. In at least one example embodiment, the seal **110** is size, for example, to extend positive pressure supply seal around a negative pressure inlet seal (e.g., openings **120**, **122**), which helps to create a barrier to dust ingress because air (a fluid) always flow from a state of high pressure to a state of low pressure.

[0065] The seal **110** may include a plurality of coupling apertures **148** that are configured to receive one or more coupling means (or fasteners) **150** that are configured to secure the seal **110** to the environment module **104**. Although the seal **110** is illustrated as including nine coupling apertures **148** configured to receive nine coupling means **150**, it should be appreciated that, in at least one example embodiment, the seal **110** may include any number of coupling apertures and the coupling apertures may be configured to receive any number of coupling means.

[0066] FIG. **5** is a rear view of the seal **110** as aligned with the cab **102**. As illustrated, the first opening **120** is aligned with a first aperture of the one or more apertures **106** in the cab **102**, the second opening **122** is aligned with a second aperture of the one or more apertures **106** in the cab **102**, and the third opening **124** of the one or more apertures **106** in the cab **102**.

[0067] While some example embodiments have been disclosed herein, it should be understood that other variations may be possible. Such variations are not to be regarded as a departure from the spirit and scope of the present disclosure, and all such modifications as would be obvious to one skilled in the art are intended to be included within the scope of the following claims.

[0068] Although described with reference to specific examples and drawings, modifications, additions, and substitutions of example embodiments may be variously made according to the description by those of ordinary skill in the art. For example, the described techniques may be performed in an order different with that of the methods described, and/or elements such as the described system, architecture, devices, circuit, and the like, may be connected or combined to be different from the above-described methods, or results may be appropriately achieved by other elements or equivalents.

1. A seal to be disposed between a structural component and an environment module, the seal comprising:
- a main body having opposing first and second sides and including one or more openings; and
- a bellow structure extending from the first side of the main body, the bellow structure including a curvature relative to a major axis of the seal.

2. The seal of claim **1**, wherein the bellow structure is a continuous bellow structure extending along a perimeter of the first side.

3. The seal of claim **2**, wherein the bellow structure is a first bellow structure, and the seal further includes:
- a second bellow structure extending around a first opening of the one or more openings.

4. The seal of claim **3**, wherein the seal further includes:
- a third bellow structure extending around a second opening of the one or more openings.

5. The seal of claim **4**, wherein the one or more openings further includes:
- a third opening and a substantially planar portion of the main body extends between a side of the third opening and the first bellow structure, a first end of the substantially planar portion having a first distance from the first bellow structure, and a second end of the substantially planar portion having a second distance from the first bellow structure, the second distance being greater than the first distance.

6. The seal of claim **1**, wherein the seal includes a metal plate defining the one or more openings, the metal plate being overmolded with a sealing material that defines the bellow structure.

7. The seal of claim **6**, wherein the sealing material has a durometer range greater than or equal to about 40 to less than or equal to about 70.

8. The seal of claim **1**, wherein the seal further includes:
- a plurality of coupling apertures extending through the main body.

9. The seal of claim **1**, wherein the seal has an average resting thickness greater than or equal to about 25 millimeters to less than or equal to about 40 millimeters.

10. A seal to be disposed between a structural component and an environment module, the seal comprising:
- a first opening surrounded or defined by a first bellow structure;
- a second opening surrounded or defined by a second bellow structure; and
- a third bellow structure surrounding the first and second bellow structures, the third bellow structure including a curvature relative to a major axis of the seal.

11. The seal of claim **10**, wherein the seal further includes:
- a third opening and a substantially planar portion of the seal extends between a side of the third opening and the first bellow structure, a first end of the substantially planar portion having a first distance from the first bellow structure, and a second end of the substantially planar portion having a second distance from the first bellow structure, the second distance being greater than the first distance.

12. The seal of claim **11**, wherein the seal further includes:
- a fourth bellow structure surrounding or defining the third opening.

13. The seal of claim **12**, wherein the seal further includes:
- a fourth opening defining a positive pressure zone.

14. The seal of claim **10**, wherein the seal includes a metal plate defining the first opening and the second opening, the metal plate being overmolded with a sealing material that defines the first, second, and third bellow structures.

15. The seal of claim **10**, wherein the seal has an average resting thickness greater than or equal to about 25 millimeters to less than or equal to about 40 millimeters.

16. A seal to be disposed between a structural component and an environment module, the seal comprising:
- one or more metal plates defining a main body having one or more openings; and

a sealing material overmolding the main body and extending from one side thereof to form a bellow structure, the bellow structure including a curvature relative to a major axis of the seal.

17. The seal of claim **16**, wherein the bellow structure extends along a perimeter of the one side of the main body, and the seal further includes:

a second bellow structure extending around a first opening of the one or more openings.

18. The seal of claim **17**, wherein the seal further includes:

a third bellow structure extending around a second opening of the one or more openings.

19. The seal of claim **16**, wherein the seal has an average resting thickness greater than or equal to about 25 millimeters to less than or equal to about 40 millimeters.

20. A system having a structural component and an environment module, the system comprising:

a seal including

a first opening disposed between a first portion of the structural component and a first portion of the environment module to create a negative pressure zone of the system; and

a second opening disposed between a second portion of the structural component and a second portion of the environment module to create a positive pressure zone of the system, the positive pressure zone surrounding the negative pressure zone.

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